TECHNICAL SPECIFICATION FOR REMOTE OPERATED LIFEBUOY

1.	**Under this contract, the successful tenderer has to comply to following Technical Specifications:-** **<u>Performance Characteristics</u>**	
	1.1	Remote Operated Lifebuoy must achieve a minimum speed between17-20 km/h on water
	1.2	Battery of the Remote Operated Lifebuoy should be capable of powering the Lifebuoy for at least 60 minutes at a minimum speed of 7.5 km/h without any load.
	1.3	Remote Operated Lifebuoy should be able to tow a person weighing a minimum of 100 kg and should be able to keep a person weighing a minimum of 200 kg afloat during the rescue.
2.	**<u>PHYSICAL CHARACTERISTICS</u>**	
	2.1	Remote Operated Lifebuoy should have a maximum weight of 18 kg
	2.2	Dimensions for Remote Operated Lifebuoy should be: Length between 100-120 cm Width between 80-85 cm Height between 20-25 cm
	2.3	Remote Operated Lifebuoy should be made of glass fibre reinforced polyester material which shall be fit for water rescue.
	2.4	Water jets should be present on both sides of Remote Operated Lifebuoy for propulsion
	2.5	Water jet inlets should be protected with grids
	2.6	Water jets should be powered by direct current (DC) brushless motors
	2.7	Remote Operated Lifebuoy should feature Rechargeable Lithium-Ion (li-ion) battery with Battery Management System (BMS).
	2.8	Remote Operated Lifebuoy should have minimum of 2 operational flashers.
	2.9	Remote Operated Lifebuoy should be in orange or red colour for better visibility.
	2.10	Remote Operated Lifebuoy should have shock-absorbing barrier at the front.
	2.11	Remote Operated Lifebuoy should have handles for individuals in the water to hold.
	2.12	Remote Operated Lifebuoy should use an RF transceiver for communication
	2.13	Remote Operated Lifebuoy should have IP68 compliant charging connector for battery charging

	2.14	Remote Operated Lifebuoy should have provision for mounting Go Pro water resistant cameras.
	2.15	Remote Operated Lifebuoy should have Manual Drive Button for manual operation
3.	**FEATURES**	
	3.1	Remote Operated Lifebuoy should turn on automatically as when as it is put on the water
	3.2	Remote Operated Lifebuoy should be capable of self-recovery from capsizing due to high waves
	3.3	Remote Operated Lifebuoy should have 2 selectable speed levels in manual mode
	3.4	Remote Operated Lifebuoy should feature GNSS satellite connection for positioning
	3.5	Remote Operated Lifebuoy should be possible to set a home position for Remote Operated Lifebuoy manually via remote control.
	3.6	Remote Operated Lifebuoy should be able to return to home position automatically in case of loss of communication
	3.7	Remote Operated Lifebuoy should be able to maintain its position automatically after returning to home position
	3.8	Remote Operated Lifebuoy should feature an easy pairing method using power button to pair with a new remote controller
	3.9	Remote Operated Lifebuoy shall have drop & go & switch on automatically when dropped in water.
	3.10	Remote Operated Lifebuoy shall flip automatically when capsized
4.	**REMOTE CONTROL**	
	4.1	Remote Operated Lifebuoy shall have a dedicated remote control system
	4.2	Remote Operated Lifebuoy remote control should have a range of at least 800 to 1000 m.
	4.3	Remote Operated Lifebuoy should be able to operate with a rechargeable battery and shall have at least 60 minutes operation time.
	4.4	Remote Operated Lifebuoy remote control should have a battery level indicator.
	4.5	Remote Operated Lifebuoy remote control should feature directional controls for forward, backward, right, and left movements.
	4.6	Remote Operated Lifebuoy remote control should have buttons to set a home position and return Remote Operated Lifebuoy to a designated home position.
5	**Documents to be submitted by the bidder along with the bid.**	
	5.1	All Relevant BIS Approval Certificate along with License number. Test Report shall

		be submitted along with the bid
	5.2	Vendor shall submit Manufacture test certificate as evidence for having used materials for various components as per specifications
	5.3	Before supply the material/goods site inspection shall be carried out at the place of OEM/Bidder by the nominated team from DG F&ES. All Expenses for inspection shall be borne by OEM/bidder.

Note-Incase of any difference between parameters given in This specification and GEM specification,Parameters given in these specification will prevail